EIDA Whitepaper v1.4

EIDA

Ephemeral Identity Distribution Architecture

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1. Abstract

Classical Public Key Infrastructure (PKI) provides static identity guarantees — it proves that a public key belongs to an entity, but cannot guarantee that a given transaction originates from that entity at that moment. This gap enables replay attacks, key reuse vulnerabilities, and long-term compromise cascades.

EIDA (Ephemeral Identity Distribution Architecture) addresses this gap by introducing identity-level forward secrecy. The core primitive, ePriPub, is a single-use ephemeral key pair derived per transaction from the user's permanent identity, which never leaves a hardware-backed isolation zone. Once used, the ePriPub components are destroyed and the CA issues a fresh single-use certificate — no persistent state is required.

EIDA shifts the trust model from centralized authority verification to cryptographic transaction uniqueness, distributing security to the user without requiring trusted intermediaries.

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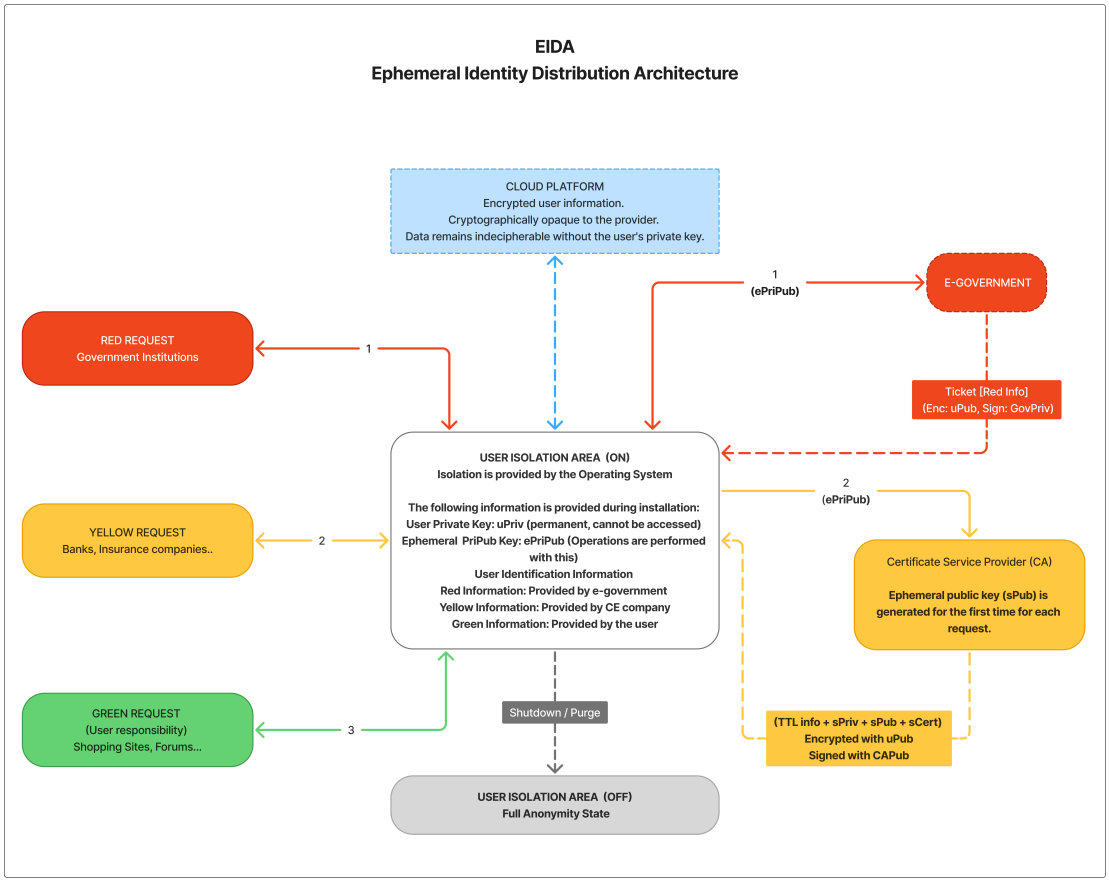
2. Problem

Today, companies responsible for securing user identities are under constant attack. A single breach of a centralized security system can expose the personal data of millions of users. This is an inevitable consequence of the current architecture: security is concentrated, and concentration creates high-value targets.

EIDA proposes distributing this responsibility to user devices. In a decentralized model, an attacker who compromises a single node gains access to only one person's data — not millions. Large technology companies are no longer the primary target. The attack surface shrinks from a single critical system to millions of individual devices.

Classical PKI reinforces this centralization problem. It guarantees that a public key belongs to an entity, but cannot verify that a specific transaction originates from that entity at that moment — enabling replay attacks, key reuse, and long-term compromise cascades.



EIDA Architecture Diagram

3. Architecture

EIDA consists of three layers:

1. User Isolation Zone

The user's permanent private key (uPriv) resides exclusively within a hardware-backed isolation zone (TPM 2.0 / TEE). It never leaves this environment under any circumstance. All cryptographic operations involving uPriv are performed inside the zone.

2. ePriPub Derivation Layer

For each transaction, a single-use ephemeral key pair is derived inside the isolation zone:

```
uPriv + nonce + timestamp → HKDF → seed → Ed25519 → (ePriv, ePub)
```

The nonce is randomly generated per transaction and combined with a timestamp, guaranteeing that no two derivations produce the same key pair. Transaction uniqueness is ensured by the (ePub, nonce, timestamp) combination — no external token coordinator is required.

3. Stateless CA Layer

The CA operates as a reactive signer — it does not scan, store, or track previous requests. Each incoming request is treated as a self-contained, atomic unit:

```
Receive: (ePub + nonce + timestamp + signature)
Verify:  signature valid? → yes
         timestamp within TTL? → yes
Generate: new (sPriv, sPub) key pair
Issue:   sCert = {sPub + expiry + metadata}, signed with CA key
Send:    (sCert + signed document) to user isolation zone
Forget:  sPriv destroyed, no state retained
```

The CA does not maintain a burned list or any persistent record of processed requests. Replay protection is provided by the short TTL window on the timestamp and the inherent single-use nature of sCert. This stateless design enables unlimited horizontal scaling — each CA node operates independently without database synchronization.

3.5 Scalability Model: Stateless Verification and the End of Revocation Lists

The Legacy Bottleneck

Traditional PKI and identity systems suffer from what can be termed the *State Exhaustion Problem*. To ensure that a token or key has not been reused or compromised, a Certificate Authority must maintain large, continuously growing Certificate Revocation Lists (CRLs) or respond to real-time status queries (OCSP). This creates a single point of latency and a database overhead that scales poorly as user populations grow into the hundreds of millions.

The EIDA Model: Zero-State Scalability

EIDA eliminates the need for a burned list or any centralized database lookup during verification. Rather than scaling the CA's state management capacity, EIDA moves the burden of security from the CA to time and hardware:

1. Hardware-Enforced Ephemerality

The ePriPub pair is generated inside a hardware-backed isolation zone (TPM/TEE) and is strictly bound to a single transaction context. The key pair ceases to exist on the client side immediately after signing — not by policy, but by design. There is nothing to revoke because there is nothing to persist.

2. Temporal Auto-Invalidation

Every EIDA token is cryptographically bound to a high-precision timestamp. The CA does not revoke keys; it simply rejects any request where `Current_Time − Timestamp > TTL`. Invalidation is a mathematical consequence of time, not a database operation.

3. O(1) Verification Complexity

The CA performs a purely mathematical verification: it checks the signature and the clock. No database is queried, no list is consulted. Verification complexity is O(1) — the operation takes the same time whether the system serves one hundred or one hundred billion users.

> **Architectural Note:** In EIDA, security is not a record maintained in a database — it is a property of the mathematics and the hardware. The CA functions as a stateless gatekeeper, making the system structurally immune to the scalability bottlenecks that constrain modern PKI deployments.

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4. ePriPub — Ephemeral Primary Public Key Primitive

ePriPub is the core primitive of EIDA. It is not a protocol — it is a single-use cryptographic identity unit that carries both signing and verification capability for exactly one transaction.

4.1 Definition

ePriPub is an ephemeral key pair (ePriv, ePub) where: - **ePriv** — ephemeral private component, used once to sign the consent token, destroyed immediately after - **ePub** — ephemeral public component, sent to CA for signature verification, discarded after use

Neither component is stored anywhere after the transaction completes.

4.2 Derivation

```
Inputs:
  uPriv     — permanent user private key (never leaves isolation zone)
  nonce     — 32-byte cryptographically random value (generated per
transaction)
  timestamp — Unix timestamp in milliseconds (generated per transaction)

Process:
  seed = HKDF(uPriv + nonce + timestamp)
  (ePriv, ePub) = Ed25519_derive(seed)

Output:
  (ePriv, ePub) — valid for this transaction only
```

4.3 Lifecycle

```
 1. DERIVE   — (ePriv, ePub) generated inside isolation zone
 2. SIGN     — consent token (ePub + nonce + timestamp) signed with ePriv
 3. TRANSMIT — (ePub + nonce + timestamp + signature) sent to CA
 4. VERIFY   — CA verifies signature and timestamp TTL
 5. DESTROY  — ePriv destroyed inside isolation zone
 6. GENERATE — CA generates new session key pair (sPriv, sPub)
 7. SIGN-DOC — CA signs document with sPriv, issues sCert
 8. RESPOND  — (sCert + signed document) sent to user isolation zone
 9. FORWARD  — isolation zone forwards to requesting institution
10. CLEAN    — sPriv destroyed by CA
```

4.4 Security Properties

- **Single-use** — sCert expires after TTL, replay attacks blocked by timestamp
- **Identity-bound** — derivation chain links ePriPub to uPriv without exposing it
- **Forward secrecy** — compromise of any ePriPub reveals nothing about uPriv or other transactions
- **Stateless CA** — CA retains no state between requests, enabling unlimited horizontal scaling

4.5 ePriPub as User Consent Proof

In classical e-signature models, the user directly signs the document with their private key. EIDA separates this into two distinct responsibilities:

```
User  → proves consent    (ePriPub)
CA    → signs document    (sPriv)
```

This separation is intentional. The CA generates a session key pair (sPriv, sPub) once per transaction and handles the actual document signature. However, the CA will only proceed with signing if a valid, unused ePriPub is present in the request.

Therefore ePriPub serves as a **cryptographic consent token**:

```
Valid ePriPub received → user consented → CA signs and issues sCert
No ePriPub             → no consent proof → CA rejects
Expired timestamp      → TTL window passed → CA rejects
```

Non-repudiation guarantee:

The user cannot later deny having initiated the transaction because: - ePriPub is derived from uPriv — only the user could have produced it - ePriPub is single-use — it cannot have been captured and replayed - (ePub + nonce + timestamp) combination is unique — it cannot be reused in another context

The CA cannot forge user consent because: - CA cannot produce a valid ePriPub without access to uPriv - uPriv never leaves the user’s isolation zone

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5. Security Analysis

5.1 Threat Model

EIDA considers the following adversaries:

- **Network attacker** — intercepts transmitted data
- **CA attacker** — compromises the certificate authority
- **Device attacker** — gains access to the user’s device
- **Replay attacker** — attempts to reuse captured transactions

5.2 Attack Scenarios

Network interception

```
Attacker captures: (ePub + nonce + timestamp + signature)
timestamp TTL → already expired (short window)
Replay attempt → CA rejects: timestamp out of TTL
uPriv → never transmitted → not exposed
Result: attack fails
```

CA compromise

```
Attacker gains access to CA
CA stores: no persistent user data, no burned list
sPriv keys → destroyed after each transaction → not recoverable
uPriv → never reaches CA → not exposed
Result: attacker finds nothing of value
```

Device compromise

```
Attacker gains access to user device
uPriv → inside TPM/TEE → cannot be extracted
Past ePriPub pairs → destroyed → not recoverable
Result: attacker is limited to future transactions
        only if physical device is held
```

Replay attack

```
Attacker captures valid (ePub + nonce + timestamp + signature)
CA checks timestamp → TTL window expired → rejected
Even within TTL: sCert already delivered and used by institution
Institution validates sCert expiry → rejected
Result: attack fails
```

5.3 What EIDA Does Not Claim

- EIDA does not protect against physical device seizure where TPM is bypassed at hardware level
- EIDA does not provide anonymity — uPriv is permanently linked to user identity
- EIDA does not address CA availability — a downed CA blocks transactions

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6. Comparison with Existing Approaches

6.1 Classical PKI

Property	Classical PKI	EIDA
Identity guarantee	Static	Per-transaction
Key reuse	Yes	Never
uPriv exposure risk	High	None
Replay protection	External	Native
CA breach impact	Critical	Minimal

Property	Classical PKI	EIDA
Forward secrecy	Session-level	Identity-level
Centralization	High	None

6.2 TLS Ephemeral (TLS 1.3)

TLS 1.3 introduces ephemeral session keys for forward secrecy. However: - Ephemeral keys apply to the **session layer**, not the identity layer - Server certificate remains static and reused across all sessions - Compromise of server certificate affects all past and future sessions

EIDA applies the ephemeral principle to the **identity layer itself** — the layer TLS never touches.

6.3 One-Time Signature Schemes (Lamport, WOTS)

One-time signature schemes share the single-use property with ePriPub. However: - They were designed for **post-quantum resistance**, not identity isolation - They do not address the centralization problem - They have no isolation zone model or stateless CA mechanism

ePriPub is architecturally distinct — its purpose is identity-level forward secrecy, not quantum resistance.

6.4 Decentralized Identity (DID, W3C)

W3C DID standard moves identity off centralized registries. However: - DID does not address per-transaction key isolation - DID keys are reused across transactions - No single-use consent token primitive exists in the DID model

EIDA is complementary to DID — ePriPub could serve as the signing primitive within a DID framework.

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7. Conclusion

The fundamental vulnerability of current internet security architecture is concentration. When identity verification depends on centralized systems, those systems become high-value targets. A single successful attack exposes millions. This is not an implementation failure — it is an architectural one.

EIDA addresses this at the architectural level by introducing three properties that classical PKI does not provide simultaneously:

- **Identity-level forward secrecy** — each transaction uses a unique, derived key pair that is destroyed after use
- **Zero uPriv exposure** — the permanent private key never leaves the hardware-backed isolation zone
- **Distributed attack surface** — there is no central repository of user identity material worth attacking

The core primitive, ePriPub, achieves these properties by combining well-established cryptographic standards — HKDF (RFC 5869) and Ed25519 — in a novel architectural pattern. No new cryptographic assumptions are required.

EIDA does not replace existing infrastructure. It extends it. CA systems, PKI chains, and existing identity frameworks remain valid — EIDA adds a per-transaction isolation layer on top of them.

The shift EIDA proposes is conceptual as much as technical: security should not be something done to users by centralized authorities. It should be something that lives with the user, on the user's device, under the user's control.

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8. References

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2. RFC 8032 - Ed25519 and Ed448 Digital Signature Algorithms
3. Bernstein, D.J. et al. (2012). High-speed high-security signatures. *Journal of Cryptographic Engineering*
4. W3C (2022). Decentralized Identifiers (DIDs) v1.0
5. Rescorla, E. (2018). The Transport Layer Security (TLS) Protocol Version 1.3, RFC 8446

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https://github.com/guvenacar/EIDA